Sigma-Cores for Lindelöf and Menger Subspaces

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Abstract

We introduce a notion of σ-core associated with a topological property P: given a space Z, a σ-P-core is, when it exists, the largest subspace of Z that can be obtained as the union of a countable increasing tower of P-subspaces. We study this notion for the Lindelöf and Menger properties.

We first prove that both properties are preserved under countable increasing unions, while neither is preserved under arbitrary increasing unions. This shows that countable towers provide the natural level of closure for the corresponding core construction.

We then analyze σ-cores in topological sums

$$Z = \bigsqcup_{i \in I} Z_i$$

whose summands are hereditarily Lindelöf, respectively hereditarily Menger. In this setting, we show that the tower-generated Lindelöf (respectively Menger) subspaces are exactly the subspaces meeting at most countably many summands. As a consequence, the σ-Lindelöf core (respectively σ-Menger core) exists if and only if the index set I is countable, in which case it coincides with Z.

Finally, we show that this countable-support description does not extend in general once a remainder is allowed: the one-point compactification of an uncountable discrete space provides a counterexample. Thus the existence of a σ-core is governed by countable support in topological sums, but fails to admit such a description in general spaces with remainder.

Keywords: General Topology; Lindelöf Spaces; Menger Spaces; Covering Properties; Topological Sums; Subspace Theory; Selection Principles; Sigma-Cores; Countable Unions; One-Point Compactification

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1. Introduction

Covering properties and their behavior under standard topological constructions have long been a central topic in general topology. In particular, the Lindelöf property and the Menger property occupy a central place in the study of covering properties and selection principles; see, for instance, Engelking [1] and Scheepers [2,3].

When a space fails to satisfy a given covering property, one may still ask whether the failure is uniform throughout the space, or whether large subspaces remain on which the property survives. This suggests the following general problem: given a topological property P and a space Z, can one identify a distinguished largest subspace of Z obtained from pieces with property P?

A first attempt would be to consider the union of all P-subspaces of Z. However, for many classical covering properties this is too coarse. In particular, while the Lindelöf and Menger properties are stable under countable increasing unions, they fail to be stable under arbitrary increasing unions. Thus, if one wants to define a largest subspace built from P-pieces, the natural level of closure is not the full union of an arbitrary family, but rather the union of a countable increasing tower.

This motivates the main definition of the present paper. Given a topological property P, we call a subspace of Z **tower-generated for** P if it is the union of a countable increasing sequence of P-subspaces. A σ-P**-core** of Z, when it exists, is the largest tower-generated P-subspace of Z.

The aim of this paper is to initiate the study of this notion for the Lindelöf and Menger properties. We do not attempt a general theory in arbitrary spaces. Instead, we focus on one concrete class where the tower-generated structure can be described explicitly, namely topological sums

$$Z = \bigsqcup_{i \in I} Z_i$$

whose summands are hereditarily Lindelöf, or hereditarily Menger. In this setting, we show that the tower-generated subspaces are exactly those meeting at most countably many summands. As a consequence, the corresponding σ-core exists if and only if the index set I is countable.

The second part of the paper shows that this countable-support description is specific to the case of topological sums. Indeed, once a remainder is adjoined, the same mechanism may fail completely: a simple compact example shows that a space of the form

$$Z = K \cup \bigsqcup_{i \in I} Z_i$$

may still be Lindelöf and Menger even when the underlying family of summands is uncountable.

The picture that emerges is therefore twofold. In topological sums of hereditary Lindelöf or hereditary Menger spaces, the existence of a σ-core is controlled entirely by countable support. By contrast, this criterion breaks down in general once a remainder is present.

The paper is organized as follows. In Section 2 we introduce P-towers, tower-generated subspaces, and σ-cores. In Section 3 we prove the basic stability of the Lindelöf and Menger properties under countable increasing unions, and show that arbitrary increasing unions are too large for the same construction. Section 4 treats σ-cores in topological sums. Section 5 provides a cautionary example showing that the countable-support description does not extend in general to spaces with remainder. We conclude in Section 6 with some final remarks and questions.

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2. P-Towers and σ-Cores

Let P be a topological property, and let Z be a topological space.

Our purpose is to formalize the idea of a largest subspace of Z built from countably many P-subspaces arranged in increasing form. Since the Lindelöf and Menger properties are stable under countable increasing unions, this provides the natural framework for the core construction studied in this paper.

Definition 2.1 (P-subspaces)

We denote by

$$\mathcal{S}_P(Z) = \{A \subseteq Z : A \text{ has property } P \text{ in the subspace topology}\}.$$

Definition 2.2 (P-tower)

A P-**tower** in Z is an increasing sequence of subspaces

$$A_0 \subseteq A_1 \subseteq A_2 \subseteq \cdots$$

such that $A_n \in \mathcal{S}_P(Z)$ for every $n < \omega$.

The associated union

$$A_\infty = \bigcup_{n<\omega} A_n$$

will be called the **subspace generated by the P-tower**.

Definition 2.3 (tower-generated P-subspace)

A subspace $B \subseteq Z$ is called **tower-generated for** P if there exists a P-tower $(A_n)_{n<\omega}$ in Z such that

$$B = \bigcup_{n<\omega} A_n .$$

We denote by

$$\mathcal{T}_P(Z)$$

the family of all tower-generated P-subspaces of Z.

Definition 2.4 (σ-P-core)

A subspace $C \subseteq Z$ is called a σ-P-**core** of Z if:

1. $C \in \mathcal{T}_P(Z)$;
2. for every $B \in \mathcal{T}_P(Z)$, one has $B \subseteq C$.

If such a subspace exists, it is unique, and we denote it by

$$C_P^\sigma(Z).$$

Thus, $C_P^\sigma(Z)$, when it exists, is the largest subspace of Z obtainable as a countable increasing union of P-subspaces.

Remark 2.5

The existence of $C_P^\sigma(Z)$ is not automatic. Even when P is stable under countable increasing unions, the family $\mathcal{T}_P(Z)$ need not contain a largest element. Sections 4 and 5 will show both positive and negative instances of this existence problem.

Proposition 2.6 (Uniqueness)

If a σ-P-core of Z exists, then it is unique.

Proof

Suppose $C, D \subseteq Z$ are both σ-P-cores. Since $C \in \mathcal{T}_P(Z)$ and D is largest in $\mathcal{T}_P(Z)$, we have $C \subseteq D$. Similarly, $D \subseteq C$. Hence $C = D$. ■

Remark 2.7

The σ-core construction is deliberately weaker than taking the union of all P-subspaces of Z. For the Lindelöf and Menger properties, arbitrary increasing unions need not preserve the property, while countable increasing unions do. This is why countable towers, rather than unrestricted families, form the basic objects of the theory.

Definition 2.8 (σ-cores in products)

If X and Y are topological spaces, a **σ-P-core of the product** $X \times Y$ means a σ-P-core of the space $X \times Y$, if it exists. It will be denoted by

$$C_P^\sigma(X, Y).$$

The main cases studied in this paper are

$$P = \text{Lindelöf} \text{ and } P = \text{Menger}.$$

Remark 2.9

In the applications below, two questions will be central:

1. which subspaces of Z are tower-generated for P;
2. whether the family $\mathcal{T}_P(Z)$ admits a largest element.

The second question is precisely the existence problem for $C_P^\sigma(Z)$.

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3. Stability under Countable Increasing Unions

The definition of σ-cores is motivated by the fact that the Lindelöf and Menger properties are preserved under countable increasing unions, while they fail to be preserved under arbitrary increasing unions. We record these facts here, since they justify the choice of countable towers as the basic objects in the core construction.

We also record the corresponding finite-union facts, which will be used repeatedly in Section 4.

Proposition 3.1

A finite union of Lindelöf subspaces is Lindelöf.

Proof

Let X and Y be Lindelöf subspaces of a space Z. Let $\mathcal{U}$ be an open cover of $X \cup Y$.

Then $\mathcal{U} \restriction X$covers X, so some countable subfamily of $\mathcal{U}$ covers X. Likewise, some countable subfamily of $\mathcal{U}$ covers Y. Their union is countable and covers $X \cup Y$. ■

Proposition 3.2

Let

$$X = \bigcup_{n<\omega} X_n ,$$

where

$$X_0 \subseteq X_1 \subseteq X_2 \subseteq \cdots$$

and each X_n is Lindelöf. Then X is Lindelöf.

Proof

Let $\mathcal{U}$ be an open cover of X. For each $n < \omega$, the family $\mathcal{U} \restriction X_n$ is an open cover of X_n. Since X_n is Lindelöf, there exists a countable subfamily $\mathcal{U}_n \subseteq \mathcal{U}$ covering X_n.

Then

$$\mathcal{V} = \bigcup_{n<\omega} \mathcal{U}_n$$

is countable and covers

$$X = \bigcup_{n<\omega} X_n .$$

Hence X is Lindelöf. ■

Proposition 3.3

A finite union of Menger subspaces is Menger.

Proof

Let X and Y be Menger subspaces of a space Z, and let $(\mathcal{U}_n)_{n<\omega}$ be a sequence of open covers of $X \cup Y$. Since X is Menger, there are finite families $\mathcal{V}_n \subseteq \mathcal{U}_n$ such that

$$X \subseteq \bigcup_{n<\omega} \cup \mathcal{V}_n.$$

Likewise, since Y is Menger, there are finite families $\mathcal{W}_n \subseteq \mathcal{U}_n$such that

$$Y \subseteq \bigcup_{n<\omega} \cup \mathcal{W}_n.$$

Then for each $n < \omega$, the family

$$\mathcal{F}_n = \mathcal{V}_n \cup \mathcal{W}_n$$

is finite and satisfies

$$X \cup Y \subseteq \bigcup_{n<\omega} \cup \mathcal{F}_n.$$

Hence $X \cup Y$ is Menger. ■

Proposition 3.4

Let

$$X = \bigcup_{n<\omega} X_n\,,$$

where

$$X_0 \subseteq X_1 \subseteq X_2 \subseteq \cdots$$

and each X_n is Menger. Then X is Menger.

Proof

Let $(\mathcal{U}_k)_{k<\omega}$ be a sequence of open covers of X. Partition ω into pairwise disjoint infinite sets:

$$\omega = \bigcup_{n<\omega} I_n\,.$$

Fix $n < \omega$. Since X_n is Menger, applied to the sequence $(\mathcal{U}_k)_{k\in I_n}$ there exist finite families

$$\mathcal{V}_k \subseteq \mathcal{U}_k (k \in I_n)$$

such that

$$X_n \subseteq \bigcup_{k\in I_n} \cup \mathcal{V}_k.$$

Doing this for each $n < \omega$, we obtain for every $k < \omega$ a finite family $\mathcal{V}_k \subseteq \mathcal{U}_k$. Then

$$X = \bigcup_{n<\omega} X_n \subseteq \bigcup_{n<\omega} \bigcup_{k \in I_n} \cup \mathcal{V}_k = \bigcup_{k<\omega} \cup \mathcal{V}_k.$$

Therefore X is Menger. ■

Corollary 3.5

Let P be either the Lindelöf property or the Menger property. Then every tower-generated P-subspace is again a P-subspace.

Proof

Immediate from Definitions 2.2 and 2.3 together with Propositions 3.2 and 3.4. ■

We now record that the corresponding stability fails for arbitrary increasing unions.

Proposition 3.6

The Lindelöf property is not preserved under arbitrary increasing unions.

Proof

Let $D = \omega_1$with the discrete topology. For each $\alpha < \omega_1$, let

$$D_\alpha = \alpha.$$

Then $(D_\alpha)_{\alpha<\omega_1}$is an increasing chain of subspaces of D, and each D_α is countable, hence Lindelöf.

However,

$$\bigcup_{\alpha<\omega_1} D_\alpha = \omega_1$$

is an uncountable discrete space, and therefore not Lindelöf. ■

Proposition 3.7

The Menger property is not preserved under arbitrary increasing unions.

Proof

With the same notation as in Proposition 3.6, each D_α is countable discrete, hence σ-compact, and therefore Menger.

On the other hand, ω_1 with the discrete topology is not Lindelöf, and every Menger space is Lindelöf. Hence ω_1 is not Menger. ■

Remark 3.8

Propositions 3.6 and 3.7 show that the union of all P-subspaces of a space is too large a construction for the Lindelöf and Menger properties. Countable increasing towers therefore provide the natural level of closure for defining a largest subspace generated by P-pieces.

Proposition 3.9

If $C_P^\sigma(Z)$exists, then

$$C_P^\sigma(Z) \in \mathcal{S}_P(Z)$$

for $P =$ Lindelöf or $P =$ Menger.

Proof

By Definition 2.4, the subspace $C_P^\sigma(Z)$, if it exists, belongs to $\mathcal{T}_P(Z)$. By Corollary 3.5, every member of $\mathcal{T}_P(Z)$ has property P. Hence $C_P^\sigma(Z) \in \mathcal{S}_P(Z)$. ■

Remark 3.10

For $P =$ Lindelöf or Menger, the existence problem for $C_P^\sigma(Z)$ is therefore not a question of preservation of P under countable towers, but a question of whether the family $\mathcal{T}_P(Z)$ admits a largest element.

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4. σ-Cores in Topological Sums

We now analyze σ-cores in topological sums, where the tower-generated structure can be described explicitly.

Let

$$Z = \bigsqcup_{i \in I} Z_i$$

be a topological sum, where each Z_i is nonempty and open in Z.

We begin with the Lindelöf case.

Lemma 4.1

If $A \subseteq Z$ is Lindelöf, then A intersects at most countably many Z_i.

Proof

For each $i \in I$, the set Z_i is open in Z, hence $A \cap Z_i$ is open in A.

Consider the family

$$\mathcal{U} = \{A \cap Z_i : i \in I, A \cap Z_i \neq \emptyset\}.$$

Since the Z_i partition Z, $\mathcal{U}$ is an open cover of A.

Because A is Lindelöf, there exists a countable subfamily

$$\{A \cap Z_{i_n} : n < \omega\}$$

covering A.

If $A \cap Z_j \neq \emptyset$, then necessarily $j = i_n$ for some $n < \omega$, since the Z_i are pairwise disjoint. Therefore

$$\{i \in I : A \cap Z_i \neq \emptyset\}$$

is countable. ■

Proposition 4.2

Assume that each Z_i is hereditarily Lindelöf. Then a subspace $A \subseteq Z$ is Lindelöf if and only if it intersects at most countably many Z_i.

Proof

(⇒) This is Lemma 4.1.

(⇐) Suppose that A intersects at most countably many Z_i, say $\{Z_{i_n} : n < \omega\}$. Set

$$B_n = \bigcup_{k \leq n} (A \cap Z_{i_k}).$$

Since each Z_{i_k} is hereditarily Lindelöf, each $A \cap Z_{i_k}$ is Lindelöf, hence B_n is Lindelöf by repeated use of Proposition 3.1. The sequence $(B_n)_{n<\omega}$ is increasing and

$$A = \bigcup_{n<\omega} B_n .$$

By Proposition 3.2, A is Lindelöf. ■

Corollary 4.3

Assume that each Z_i is hereditarily Lindelöf. Then the tower-generated Lindelöf subspaces of Z are exactly its Lindelöf subspaces.

Proof

If $A \subseteq Z$ is tower-generated Lindelöf, then A is Lindelöf by Corollary 3.5.

Conversely, let $A \subseteq Z$ be Lindelöf. By Proposition 4.2, Aintersects at most countably many summands, say $\{Z_{i_n} : n < \omega\}$. Define

$$B_n = \bigcup_{k \leq n} (A \cap Z_{i_k}).$$

As above, each B_n is Lindelöf, the sequence is increasing, and

$$A = \bigcup_{n<\omega} B_n .$$

Thus A is tower-generated Lindelöf. ■

Theorem 4.4

Assume that each Z_i is nonempty and hereditarily Lindelöf. Then the σ-Lindelöf core $C_{\mathrm{L}}^{\sigma}(Z)$ exists if and only if I is countable. In that case,

$$C_{\mathrm{L}}^{\sigma}(Z) = Z.$$

Proof

Suppose first that I is countable. Then Z intersects at most countably many summands, so Z is Lindelöf by Proposition 4.2. Hence Z is tower-generated Lindelöf, and therefore

$$C_{\mathrm{L}}^{\sigma}(Z) = Z.$$

Suppose now that I is uncountable. By Corollary 4.3, the tower-generated Lindelöf subspaces of Z are exactly the Lindelöf subspaces of Z, hence exactly the subspaces intersecting at most countably many summands.

Let $A \subseteq Z$ be any tower-generated Lindelöf subspace. Then A intersects only countably many Z_i. Since I is uncountable, there exists $j \in I$ such that

$$A \cap Z_j = \emptyset.$$

Choose a point $x_j \in Z_j$. Then

$$A' = A \cup \{x_j\}$$

still intersects at most countably many summands, so A' is Lindelöf by Proposition 4.2, and hence tower-generated Lindelöf by Corollary 4.3.

Since $A \subsetneq A'$, no tower-generated Lindelöf subspace can be largest. Therefore $C_{\mathrm{L}}^{\sigma}(Z)$ does not exist. ■

The Menger case is parallel.

Proposition 4.5

Assume that each Z_i is hereditarily Menger. Then a subspace $A \subseteq Z$ is Menger if and only if it intersects at most countably many Z_i.

Proof

(⇒) Every Menger space is Lindelöf, so Lemma 4.1 applies.

(⇐) Suppose that A intersects at most countably many Z_i, say $\{Z_{i_n} : n < \omega\}$. Set

$$B_n = \bigcup_{k \le n} (A \cap Z_{i_k}).$$

Since each Z_{i_k} is hereditarily Menger, each $A \cap Z_{i_k}$ is Menger, hence B_n is Menger by repeated use of Proposition 3.3. The sequence $(B_n)_{n<\omega}$ is increasing and

$$A = \bigcup_{n<\omega} B_n .$$

By Proposition 3.4, A is Menger. ■

Corollary 4.6

Assume that each Z_i is hereditarily Menger. Then the tower-generated Menger subspaces of Z are exactly its Menger subspaces.

Proof

As in Corollary 4.3, using Proposition 4.5 and Proposition 3.4. ■

Theorem 4.7

Assume that each Z_i is nonempty and hereditarily Menger. Then the σ-Menger core $C_{\mathrm{M}}^{\sigma}(Z)$ exists if and only if I is countable. In that case,

$$C_{\mathrm{M}}^{\sigma}(Z) = Z.$$

Proof

The proof is identical to that of Theorem 4.4, replacing Lindelöf by Menger and using Proposition 4.5 and Corollary 4.6. ■

Remark 4.8

In topological sums of hereditary Lindelöf or hereditary Menger spaces, the existence of a σ-core is governed entirely by countable support. The obstruction in the uncountable case is not the failure of the covering property on the individual summands, but the absence of any largest countable choice of summands.

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5. A Cautionary Example with a Remainder

The countable-support description obtained in Section 4 might suggest that the same mechanism should extend to spaces of the form

$$Z = K \cup \bigsqcup_{i \in I} Z_i,$$

where K is a hereditary remainder and the family $\{Z_i : i \in I\}$ is pairwise disjoint and open. The following example shows that this is false in general, even in a very simple compact case.

Example 5.1

Let D be an uncountable discrete space, and let

$$\alpha D = D \cup \{\infty\}$$

be its one-point compactification; see [1].

Then

$$\alpha D = \{\infty\} \cup \bigsqcup_{d \in D} \{d\},$$

where each singleton $\{d\}$ is clopen, hence hereditarily compact, hereditarily Lindelöf, and hereditarily Menger, and the remainder $\{\infty\}$ has the same hereditary properties.

However, αD itself is compact. Hence it is both Lindelöf and Menger.

Proposition 5.2

With the notation above,

$$C_{\mathrm{L}}^{\sigma}(\alpha D) = \alpha D \text{ and } C_{\mathrm{M}}^{\sigma}(\alpha D) = \alpha D.$$

Proof

Since αD is compact, it is Lindelöf and Menger. Therefore it is tower-generated for both properties, via the constant tower

$$\alpha D \subseteq \alpha D \subseteq \alpha D \subseteq \cdots .$$

As the whole space, it is necessarily the largest such subspace. ■

Remark 5.3

Proposition 5.2 shows that the criterion from Section 4 does not extend to arbitrary spaces with remainder. In particular, the existence of a σ-Lindelöf or σ-Menger core is no longer governed solely by countable support once a remainder is present.

The point is that the remainder may compactify or otherwise reorganize the uncountable family of summands, so that the whole space acquires the relevant covering property even when the underlying family of summands is uncountable.

Question 5.4

It would be interesting to determine natural additional hypotheses on a decomposition

$$Z = K \cup \bigsqcup_{i \in I} Z_i$$

under which a countable-support description of $C_{\mathrm{L}}^{\sigma}(Z)$ or $C_{\mathrm{M}}^{\sigma}(Z)$ can still be recovered.

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6. Final Remarks and Questions

In this paper we introduced the notion of a σ-core associated with a topological property P, defined as the largest subspace generated by a countable increasing tower of P-subspaces, whenever such a largest element exists. We studied this notion for the Lindelöf and Menger properties.

The basic structural point is that both properties are preserved under countable increasing unions, but not under arbitrary increasing unions. This makes countable towers the natural closure mechanism for the corresponding core construction.

For topological sums of hereditarily Lindelöf spaces, and likewise for topological sums of hereditarily Menger spaces, we showed that the tower-generated subspaces are exactly those meeting at most countably many summands. As a consequence, the corresponding σ-core exists exactly in the countable case.

We also showed that this countable-support description does not extend in general beyond topological sums. The one-point compactification of an uncountable discrete space provides a simple counterexample: once a remainder is introduced, the whole space may satisfy the relevant covering property even when the family of summands is uncountable.

These results suggest several natural questions.

1. For which classes of spaces beyond topological sums can tower-generated Lindelöf or Menger subspaces be described explicitly?
2. Under what additional assumptions on a decomposition

$$Z = K \cup \bigsqcup_{i \in I} Z_i$$

   can a countable-support description of $C_{\mathrm{L}}^{\sigma}(Z)$ or $C_{\mathrm{M}}^{\sigma}(Z)$ still be recovered?
3. Are there natural conditions characterizing spaces Z for which $C_{\mathrm{L}}^{\sigma}(Z)$ or $C_{\mathrm{M}}^{\sigma}(Z)$ exists?
4. To what extent can analogous constructions be developed for other covering properties, such as the Hurewicz property or related selection principles?

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